

Reactive Oxygen Detoxification Contributes to *Mycobacterium abscessus* Antibiotic Survival

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eLife Assessment

Using a transposon sequencing (TN-seq) approach, the authors identified key genetic determinants of drug tolerance in *Mycobacterium abscessus*. Given that *M. abscessus* is inherently resistant to multiple antibiotics, this **valuable** study makes a significant contribution by uncovering how antibiotic tolerance is linked to reactive oxygen species (ROS) in this non-tuberculous mycobacterial (NTM) species. The **solid** findings further strengthen the growing evidence that ROS play a central role in the mechanism of antibiotic action and tolerance in mycobacteria. However, the manuscript would benefit from improved clarity of presentation and corrections in the reference section.

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Abstract

When a population of bacteria are exposed to a bactericidal antibiotic most cells die rapidly. However, a sub-population of antibiotic-tolerant cells known as “persister cells” can survive for prolonged periods, and antibiotic tolerance can be strongly induced in a population by stresses such as nutrient deprivation. However, the pathways required to maintain viability in this setting, and how they are regulated are poorly understood. To identify genetic determinants of antibiotic tolerance in mycobacteria, we carried out transposon mutagenesis insertion sequencing (Tn-Seq) screens in *Mycobacterium abscessus* (*Mabs*) exposed to bactericidal translation-inhibiting antibiotics. This analysis identified genes essential for the survival of both spontaneous persister cells as well as for stress-induced tolerance, allowing the first genetic comparison of these states in mycobacteria. Pathway analysis identified multiple genes involved in the detoxification of reactive oxygen species (ROS), including the catalase-peroxidase *katG*, which contributed to survival in both stressed and un-stressed populations. In addition, we found that endogenous ROS were generated following antibiotic exposure, and that hypoxia significantly impaired bacterial killing. Thus, in *Mabs*, the

lethality of some antibiotics is amplified by toxic ROS accumulation, and antibiotic-tolerant cells require detoxification systems to remain viable.

Introduction

A key goal of antibiotic therapy is, in conjunction with the immune system, to eradicate the infecting bacteria. While many common bacterial infections respond rapidly to antibiotics, with 1–2 weeks of therapy sufficient to achieve high cure rates, there are also infections where bacterial clearance is slow and often incomplete^{1,2}. This challenge is exemplified by mycobacterial infections. Fully-susceptible *Mtb* requires multiple antibiotics for four months or longer^{3,4}, and infections by non-tuberculous mycobacteria (NTM) such as *Mycobacterium abscessus* (*Mabs*) are often even more difficult to eradicate; *Mabs* often requires treatment for 12–18 months, and even then has a 50% relapse rate^{5,6}.

While the ability of mycobacteria to escape antibiotic-mediated killing is multifactorial, the phenomenon of antibiotic tolerance is likely an important contributor^{7–10}. Studies dating from the 1940s noted that when a population of susceptible bacteria were exposed to a bactericidal antibiotic such as penicillin, the majority of the population died within a few hours, but that a small sub-population of cells remained viable for days¹¹. Importantly, these antibiotic-tolerant cells referred to as “persister cells” had not acquired a mutation conferring heritable antibiotic resistance, and do not grow in the presence of the antibiotic. Rather, they had entered into a readily-reversible phenotypic state where, despite antibiotic-mediated inhibition of critical processes, they are able to survive^{12–14}. In addition, a number of physiologic stresses increase antibiotic tolerance in a population, as bacterial cell death is markedly slowed by stresses such as nutrient deprivation or acidic pH^{10,11,15,16}. Notably, these same stresses are encountered in the lysosome of an activated immune cell¹⁷, and studies of pathogens isolated from activated macrophages indeed show a strong immune-mediated increase antibiotic tolerance^{9,18}. Thus, paradoxically, the immune system can actually impede bacterial eradication by antibiotics.

Antibiotic tolerance has been studied extensively in model systems such as *Escherichia coli*, which has provided important insights, but also highlighted uncertainties of current models. Several different regulators of antibiotic tolerance have been identified in *E. coli* including the HipBA toxin-antitoxin system¹⁹, guanosine pentaphosphate ((p)ppGpp) synthesis by RelA/SpoT enzymes^{20,21}, and the activity of Lon protease²⁰. In each of these models, the postulated mechanism is to arrest cell division and render the process targeted by antibiotics non-essential. However, important questions remain unanswered. It is unclear how cells remain viable when critical processes such as transcription or translation are blocked by antibiotics, as well as how the process is regulated and induced by stress. Even the mechanism of cell death following antibiotic exposure itself has been a matter of debate - originally antibiotics were presumed to kill bacteria as a direct result of inhibition of their target molecule, such as β -lactam antibiotics disrupting cell wall integrity, directly leading to mechanical cell lysis²². However, a number of studies, largely from *E. coli*, have demonstrated that in addition to the initial target inhibition, which may be bacteriostatic, bactericidal antibiotics can also cause secondary lethal ROS accumulation, leading to cell death^{14,23–28}.

Mycobacterial persister cells are particularly resilient, as *Mycobacterium smegmatis* (*Msmeg*) and *Mycobacterium tuberculosis* (*Mtb*) persisters can endure many weeks of antibiotic exposure, and stress-induced antibiotic tolerance develops readily^{10,14,29}. *Mabs* is a species of rapidly-growing mycobacteria, with a doubling time of ~4h in rich media³⁰, and while often environmental, it causes opportunistic infections in patients with structural lung disease such as cystic fibrosis. It is also among the most difficult of all bacterial pathogens to treat, because in addition to forming persister cells, it is also intrinsically resistant to many classes of antibiotics,

leaving few treatment options⁵. This leads to the use of antibiotics with greater toxicity to patients, and a need to use these agents for prolonged periods to prevent relapse. Thus, identifying the genes that *Mabs* persister cells rely on for survival, as well as the genes involved in the induction and maintenance of stress-induced antibiotic tolerance might highlight pathways that could be targeted therapeutically to eliminate antibiotic-tolerant cells.

Previous genetic screens have studied antibiotic responses in mycobacteria, with some evaluating heritable resistance, and others investigating tolerance. Several studies of resistance have successfully used either transposon mutagenesis with insertion site sequencing (Tn-Seq) or CRISPR-based transcriptional repression with high-throughput sequencing of guide RNAs (CRISPRi) to identify genes promoting growth in sub-inhibitory concentrations of antibiotic. These studies have provided insights such as highlighting the importance of cell membrane permeability as a mechanism controlling antibiotic penetration into the cytoplasm^{31–33}.

Persister formation has proven challenging to study, likely because the low frequency of persisters leads to population bottlenecks that confound genetic analysis. Although screens in *Mtb* have been conducted in macrophages and mice, and genes such as *glpK* and *cinA* identified, overall the number of mutants isolated in these screens has been low^{34–35}. There has been one effective in vitro Tn-Seq study examining antibiotic tolerance in *Mtb* exposed to starvation and rifampin that isolated over 100 mutants²⁷, demonstrating the feasibility of genetic screening in this context. However, whether these phenotypes seen with rifampin in *Mtb* extend to other mycobacteria and other antibiotics remain to be determined.

Here, we study antibiotic tolerance in *Mabs* and describe the results of a genome-wide Tn-seq screens seeking to identify the genes required for both spontaneous persister cell survival as well as starvation-induced antibiotic tolerance following exposure to translation-inhibiting antibiotics. We identified several discrete processes contributing to survival and observed a prominent role for ROS detoxifying factors such as the catalase-peroxidase enzyme KatG, which contributed to both spontaneous persister survival and starvation-induced antibiotic tolerance. Consistent with the protection conferred by KatG, we found that endogenous ROS accumulated following antibiotic exposure, and that the removal of oxygen significantly impaired bacterial killing. Taken together these findings support a model in mycobacteria where the lethality of translation-inhibiting antibiotics is amplified by a secondary accumulation of toxic ROS, and that survival requires active detoxification systems.

Results

Starvation-induced antibiotic tolerance in mycobacteria

We first sought to develop conditions suitable for genetic analysis of persister cell survival and stress-induced antibiotic tolerance mycobacteria. Genetic screens examining persister cell physiology face two inherent obstacles. First, these cells are rare in unstressed bacterial populations, and antibiotic-mediated cell death creates population bottlenecks that obscure mutant phenotypes. Second, most mycobacterial populations contain spontaneous drug-resistant mutants that can expand if the population is exposed to a single antibiotic. To overcome these obstacles, we sought to establish large-scale, high-density culture conditions to prevent genetic bottlenecks, and used multiple antibiotics to suppress spontaneous drug-resistant mutants. We began by assessing the feasibility of this approach using wild-type *Msmeg*. We exposed the cells to either the combination of rifampin, isoniazid, and ethambutol (RIF/INH/EMB) used to treat *Mtb*, or to the combination of tigecycline and linezolid (TIG/LZD), two translation-inhibiting antibiotics frequently used to treat *Mabs*^{6,36}, and empirically determined the minimum inhibitory concentrations (MICs) and minimum bactericidal concentrations (MBC) for each antibiotic under the high-density culture conditions that would be needed for genetic analysis of persister cells.

Both antibiotic combinations reduced the bacterial population >1000-fold within 72h (**Figure 1A**). We then evaluated both spontaneous persister formation and stress-induced tolerance under these conditions in *Msmeg*. We compared logarithmically growing (mid-log) cultures in 7H9 rich media to cultures starved for 2 days in phosphate buffered saline (PBS) prior to addition of antibiotics. Consistent with expectations, we found a marked increase in antibiotic tolerance in starved cultures, with a 100-fold increase in survival following TIG/LZD exposure and a 10,000-fold increase following RIF/INH/EMB exposure (**Figure 1A**).

We next examined two species of pathogenic mycobacteria under these conditions to similarly assess starvation-induced antibiotic tolerance. We again compared cells starved in PBS to logarithmically growing cells in 7H9 and found that cultures of wild-type *Mabs* (ATCC 19977) and *Mtb* (Erdman) also displayed dramatic increases antibiotic tolerance in nutrient-deprived cultures (**Figure 1B,C**). Notably, for *Mabs* and *Msmeg*, the development of tolerance required an adaptation period of several days under starvation conditions, as survival was dramatically impaired if cells were shifted immediately into nutrient-deficient conditions with antibiotics, suggesting that a regulated process needed to be completed. Surprisingly, *Mtb* tolerance developed rapidly without pre-adaptation suggesting that this organism might have additional response pathways enabling more rapid adaptation.

Identification of pathways needed for antibiotic tolerance in *Mabs*

We used these conditions to carry out Tn-Seq screens in *Mabs* to identify genes necessary for both the survival of spontaneous persister cells and for starvation-induced antibiotic tolerance. We conducted the screen using a *Mabs* *Himar1* Tn library comprised of ~55,000 mutations across ~91,000 possible TA insertion sites covering all 4,992 non-essential *Mabs* genes in strain ATCC 19977. To study spontaneous persister cells, cultures were maintained in continuous log-phase in 7H9 rich media for 48h prior to antibiotic exposure and then exposed to TIG/LZD for 6 days (a point at which spontaneous persister cells comprise the majority of the population) a point at which spontaneous persister cells comprise the majority of the population. To study starvation-induced tolerance, cultures were starved in PBS for 48h prior to antibiotic exposure in PBS. Following antibiotic treatment, cells were then washed and resuspended in antibiotic-free liquid media to recover and passaged 1:100 three times in continuous log-phase to expand surviving cells. We then isolated genomic DNA, sequenced the Tn insertion sites, and used TRANSIT software to quantify the abundance of each Tn mutant across different conditions to identify genes with statistically-significant differences in distribution. We identified 277 *Mabs* genes required for surviving TIG/LZD exposure in rich media, 271 genes required for survival during starvation and 362 genes required to survive the combined exposure to antibiotics and starvation (Log₂ fold-change > 0.5 and Benjamini–Hochberg adjusted p-value (p-adj.) ≤ 0.05; **Figure 2B-E**). Of the genes required for survival, ~60% were required in both nutrient-replete and starvation states, although condition-specific determinants were also seen. As expected, we identified genes with already-established functions in antibiotic responses, including *MAB_2752* and *MAB_2753* which are both homologs of known antibiotic transporters in *Mtb*, as well as tetracycline-responsive transcription factors like *MAB_4687* and *MAB_0314c* (**Table S1**), indicating an ability of these Tn-Seq conditions to identify physiologically relevant genes known to mitigate antibiotic stress.

To identify other cellular processes necessary for survival we performed pathway enrichment analysis on the set of genes identified by Tn-Seq. We used the DAVID analysis tool to perform systematic queries of the KEGG, GO, and Uniprot databases and identify over-represented processes and pathways. Interestingly, although cells were exposed to translation-inhibiting antibiotics, and no exogenous oxidative or nitrosative stress was applied, we identified a number of factors needed to combat these stresses in spontaneous persister cells. This included *bfrB* (bactoferritin), *ahpE* (peroxiredoxin) and *katG* (catalase/peroxidase) as well as 5 components of the bacterial proteasome pathway, known to mediate resistance to nitrosative stress in *Mtb*. We also identified multiple members of DNA-damage response pathways including *recF*, *recG*, *uvrA*, *uvrB* and *uvrC*. Examining genes required for starvation-

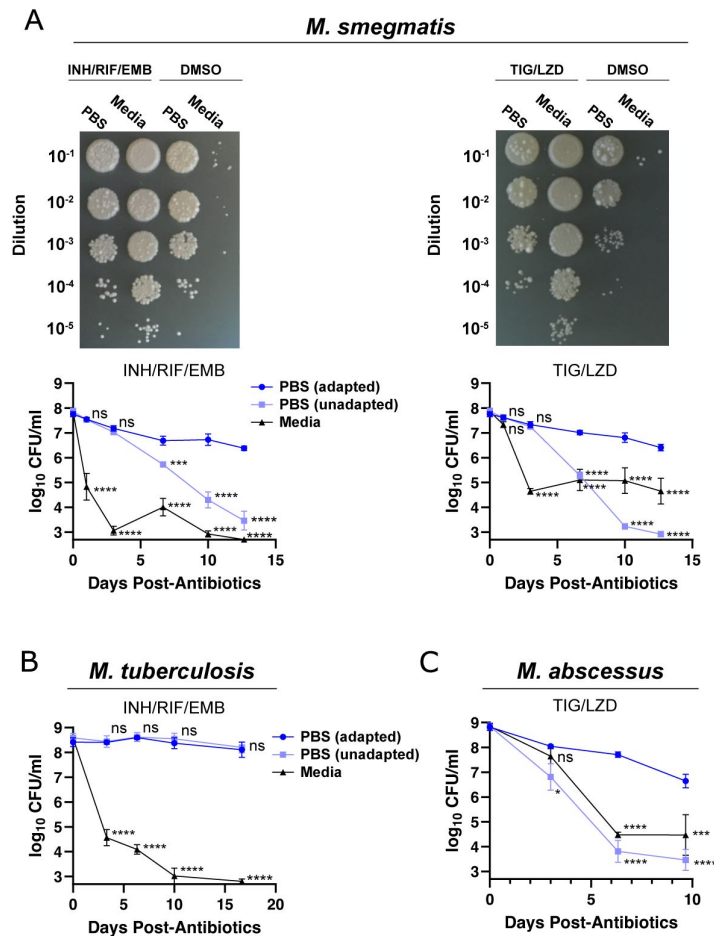


Figure 1.

Starvation induces antibiotic tolerance in diverse mycobacteria.

(A) *Msmeg*, (B) *Mtb* or (C) *Mabs* were grown in 7H9 rich media or starved in PBS prior to the addition of antibiotics and surviving colony forming units (CFU) enumerated. For the rapidly growing mycobacteria, *Msmeg* and *Mabs*, cells were allowed to adapt for 48h prior to antibiotics, for slow-growing *Mtb*, cells were allowed to adapt for 14-21d prior to antibiotics. Samples without pre-adaptation were washed and placed directly into PBS with antibiotics. Antibiotic concentrations were: *Msmeg* - Isoniazid (INH) 32 μ g/ml (8 x MIC), rifampin (RIF) 32 μ g/ml (8 x MIC), ethambutol (EMB) 4 μ g/ml (8 x MIC), tigecycline (TIG) 1.25 μ g/ml (8 x MIC), linezolid (LZD) 2.5 μ g/ml (8 x MIC). *Mtb* - RIF 0.1 μ g/ml (4 x MIC), INH at 0.1 μ g/ml (4 x MIC), EMB at 8 μ g/ml (4 x MIC). *Mabs* - TIG 10 μ g/ml (8 x MIC), LZD 100 μ g/ml (20 x MIC). Antibiotics with half-lives shorter than the duration of experiment were re-added at the following intervals: TIG, EMB every 3d; RIF, INH every 6d. Error bars represent SEM, statistical significance is calculated at each time point using student's t test. *****: $p < 0.0001$, ***: $p < 0.001$, **: $p < 0.01$, *: $p < 0.05$, ns: $p > 0.05$. Data are combined from 3 independent experiments.

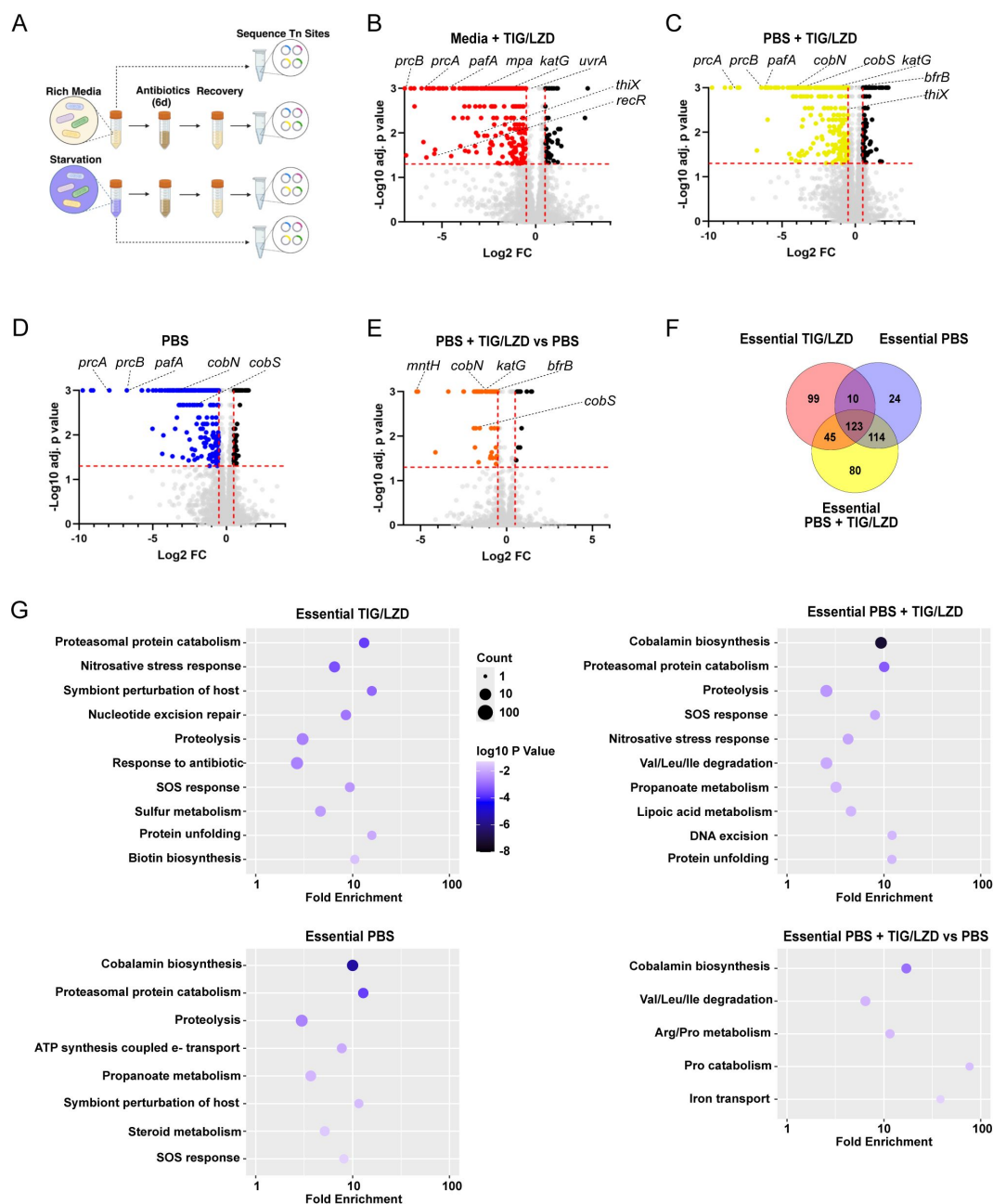


Figure 2.

Tn-Seq identifies genes required for antibiotic tolerance in *Mabs*.

(A) Experimental design. (B-E) Tn-Seq analysis showing relative abundance of individual genes under the indicated conditions. Genes depleted relative to the input population have negative values. All cultures were fully aerated throughout the experiment and cultures without antibiotics received an equal volume of DMSO. For (B-D) gene abundance in the indicated condition is measured relative to the input log-phase population. In (E) an additional comparison is made for PBS with antibiotics relative to the PBS condition. Genes with significant decreases in abundance are shown in color (p -adj. < 0.05 and $\log_2$ fold-change > 0.5) using the Benjamini-Hochberg adjustment for multiple hypothesis testing. (F) Number of genes essential in each condition relative to the input population. (G) Pathway enrichment analysis of the essential genes in each condition using the DAVID knowledgebase ($p < 0.05$). Screens were run as 3 independent experiments and the combined results analyzed. Antibiotic conditions were as described above.

induced tolerance, a number of the same pathways were again seen, and the mutant with the greatest survival defect in this context was *mntH*, a redox-regulated Mn^{2+}/Zn^{2+} transporter implicated in peroxide resistance in other organisms^{44–45}.

To independently confirm a role in antibiotic tolerance for a set of genes from diverse pathways that were identified by Tn-Seq we selected a set of genes required for survival, representing several of the functional pathways identified, and used oligonucleotide-mediated recombineering (ORBIT)⁴⁶ to disrupt their open reading frames. The initial genes selected were *pafA* (proteasome pathway), *katG* (catalase-peroxidase), *recR* (DNA repair), *blaR* (β -lactam sensing), and *MAB_1456c* (cobalamin synthesis). To control for non-specific effects of antibiotic selection during the recombineering process, we created a control strain using ORBIT to target a non-coding intergenic region downstream of a redundant tRNA gene (*MAB_t5030c*). We then individually screened each of these mutants to determine if they displayed deficits in survival by exposing cells to TIG/LZD, either in rich 7H9 media or under starvation conditions, as had been done in the Tn-Seq screen. For $\Delta katG$ we detected clear defects in survival as soon as 3 days after antibiotic exposure in either rich media or under starvation conditions, corroborating the results of our Tn-Seq analysis (**Figure 3A**). We observed similar, albeit smaller, defects in the $\Delta pafA$, ΔMAB_1456c , and $\Delta blaR$, mutants under the conditions predicted by the screen, whereas we saw no defect in the $\Delta recR$ mutant (**Figure 3B–E**). We also noted that several mutants displayed delays in growth recovery after antibiotic removal at times before survival defects were noted (**Figure 3 Supplement**). To further confirm the role of *katG* and *pafA*, and exclude off-target effects of recombineering, we performed genetic complementation analysis by restoring a wild-type copy of each gene into the respective $\Delta pafA$ and $\Delta katG$ mutants. In each case, we integrated a single copy of the wild-type gene, under the control of its endogenous promoter, into the genome at the L5 *attB* site (hereafter *pafA*⁺, *katG*⁺ strains), and constructed isogenic control strains with an empty vector integrated at the same site (hereafter *pafA*[−], *katG*[−] strains). We confirmed expression of the re-introduced copy of each gene by RT-qPCR in the *pafA*⁺, and *katG*⁺ strains, and found expression within roughly 2-fold of endogenous wild-type levels (**Figure 4A,D**). We then challenged these strains with TIG/LZD as before. In rich media, where the $\Delta katG$ mutants have a moderate survival defect, the *katG*⁺ strain had roughly a 50-fold increase in viable cells relative to the *katG*[−] strain. We then exposed cells to antibiotics under starvation conditions, where the $\Delta katG$ mutant phenotype is more severe. Under these conditions the *katG*[−] cells succumbed rapidly between 3d and 10d after antibiotic exposure, with a 1,000-fold decrease in viable cells relative to control cells, whereas the *katG*⁺ strain showed a near-complete restoration of antibiotic tolerance (**Figure 4B**). We analogously examined complementation of $\Delta pafA$ mutants, and although the phenotype of the $\Delta pafA$ mutant is less severe overall than a $\Delta katG$ mutant, we saw a similar restoration of survival in *pafA*⁺ cells relative to *pafA*[−] cells (**Figure 4E**). We next evaluated whether the *pafA*[−], and *katG*[−] strains were overall more sensitive to the growth inhibitory effects of TIG/LZD, or whether they had specific defects in survival above the mean bactericidal concentration. We performed MIC determination for TIG and LZD individually for each strain, comparing the *katG*⁺/*katG*[−] and *pafA*⁺/*pafA*[−] strains. We found that that the MICs for each of these strains were unchanged, demonstrating that these mutants were not more readily inhibited by these antibiotics. Instead, they have more rapid kinetics of cell death at bactericidal concentrations, consistent with a specific defect in antibiotic tolerance, and supporting a model whereby an initial growth-arresting inhibition of the direct antibiotic target can be uncoupled mechanistically from a distinct cell-death step (**Figure 4C,F**), as has been seen with other antibiotic classes²⁶.

Reactive oxygen contributes to antibiotic lethality in *Mabs*

We next investigated the role of KatG and reactive oxygen in antibiotic tolerance more broadly. We began by assessing whether KatG conferred protection from other antibiotics with diverse mechanisms of action, selecting antibiotics that are used clinically for mycobacterial infections. Because *katG*[−] mutants showed the greatest defects in starvation-induced tolerance, we analyzed survival of *katG*⁺ and *katG*[−] strains in starvation-adapted cultures exposed to a panel of different

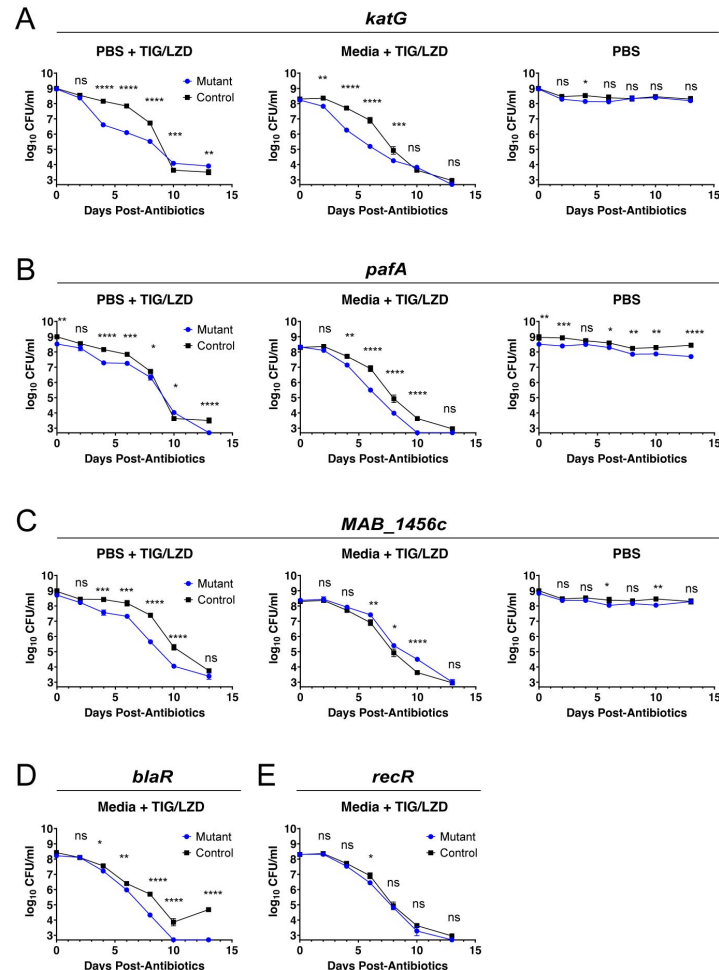


Figure 3.

Validation of Tn-Seq results.

(A-E) Homologous recombination was used to delete the indicated genes, or to generate a control strain targeting a distant intergenic region distal to the non-essential tRNA gene *MAB_t5030c*. Each strain was either grown in 7H9 rich media or starved in PBS for 48 prior to the addition of antibiotics as indicated. The conditions tested here correspond to the conditions in the Tn-Seq analysis where a phenotype was observed. Data from additional conditions are in [Figure 3 Supplemental](#). Comparisons in panels A-C are made to the same control strain but plotted independently for clarity. Error bars represent SEM, statistical significance is calculated at each time point using student's t test. ****: $p < 0.0001$, ***: $p < 0.001$, **: $p < 0.01$, *: $p < 0.05$, ns: $p > 0.05$. Antibiotics were added as described above. Data are representative of 4 independent experiments.

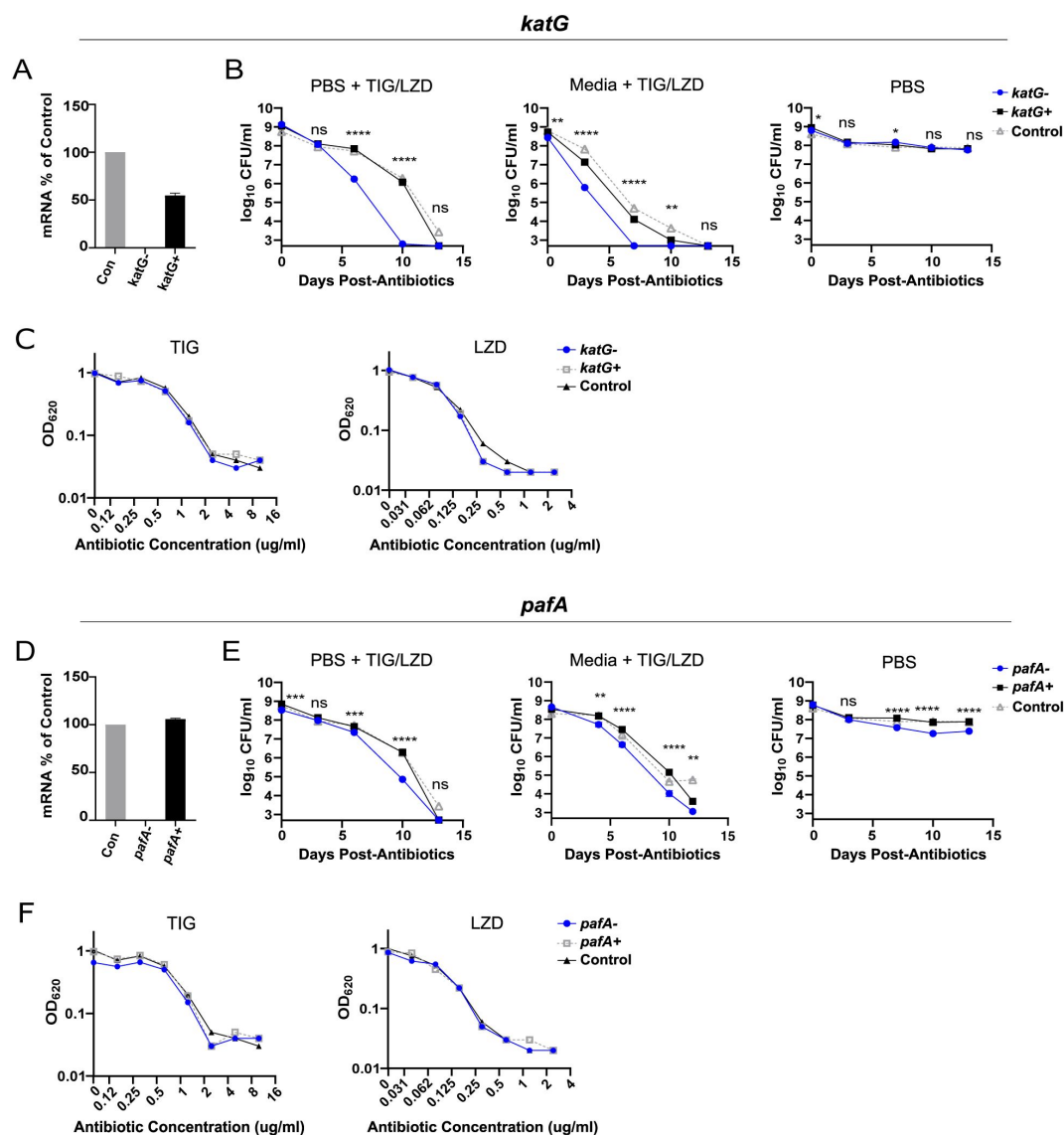


Figure 4.

Complementation analysis of *katG* and *pafA* mutants confirms their role in antibiotic tolerance.

(A) RT-qPCR analysis of *katG* expression in *katG*⁻ ($\Delta katG::pmv306$), *katG*⁺ ($\Delta katG::pmv306 katG$), and control strain (ORBIT intergenic::pmv306). (B) CFU over time for *katG*⁺/*katG*⁻ strains. (C) MICs for *katG*⁺/*katG*⁻ strains. (D) Expression of *pafA* in *pafA*⁻ ($\Delta pafA::pmv306$), *pafA*⁺ (*pafA::pmv306 pafA*) and control strain. (E) CFU over time for *pafA*⁺/*pafA*⁻ strains. (F) MICs for *pafA*⁺/*pafA*⁻ strains. Antibiotic concentrations in (A-B, D-E) are as described above. Error bars represent SEM, statistical significance is calculated at each time point using student's t test between *katG*⁺/*katG*⁻ strains in (B) and between *pafA*⁺/*pafA*⁻ strains in (E). ****: $p < 0.0001$, ***: $p < 0.001$, **: $p < 0.01$, *: $p < 0.05$, ns: $p > 0.05$. Antibiotics were added as described above.

antibiotics. Because both TIG and LZD both act by inhibiting translation, we began by exposing cells to either TIG or LZD alone. As expected, the degree of bacterial killing was significantly less with either agent alone than when they are added in combination. Upon exposure to either of these antibiotics the *katG*⁻ cells died more rapidly than *katG*⁺ cells, though we note that the final proportion of persister cells in the population was unchanged in *katG*⁻ cells (**Figure 5A** [↗](#)). When we exposed cells to rifabutin, an RNA polymerase inhibitor, we saw a similar effect, with a 100-fold loss of viability in *katG*⁻ cells relative to the *katG*⁺ cells (**Figure 5B** [↗](#)). In contrast, when we exposed cultures to either levofloxacin (topoisomerase inhibitor) or cefoxitin (β-lactam inhibitor of peptidoglycan cross-linking), *katG* had little to no effect on cell viability (**Figure 5C-D** [↗](#)). Thus, the role of KatG is context-dependent, suggesting that some antibiotics generate oxidative stress that can be ameliorated by KatG while others do not.

The identification of *katG* as essential for cells to survive exposure to TIG/LZD suggests that ROS are present and causing damage. Although TIG/LZD are translation inhibitors that do not directly generate ROS we evaluated whether they might nonetheless be triggering ROS accumulation as a secondary effect. We examined ROS levels in control *Mabs* using the ROS indicator dye cellROX, that is retained in cells when it becomes oxidized⁴⁷[↗](#). At baseline, during log-phase growth in rich media < 2% of cells had ROS accumulation (**Figure 5E** [↗](#)). We saw a moderate increase in ROS accumulation in starved cultures, with roughly 7% of the population cellROX⁺. However, when cells were exposed to antibiotics we saw a dramatic accumulation of ROS, with 57% of cells becoming cellROX⁺ when exposed to TIG/LZD in rich media and 33% of PBS-starved cells becoming cellROX⁺ when exposed to TIG/LZD. Taken together, these data indicate that translation inhibition does indeed have important downstream effects on cellular redox balance, with ROS accumulation that could be contributing to the lethal effects of antibiotics.

We next tested whether ROS were contributing to cell death by reducing ROS production and assessing the impact on cell viability. A well-established system for studying hypoxia in mycobacteria is the Wayne Model of gradual-onset hypoxia, whereby low-density cultures are inoculated in sealed vessels with minimal headspace. As the culture slowly grows, the soluble oxygen is consumed, resulting in the slow onset of hypoxia over several days, a process that can be monitored by the decolorization of Methylene Blue dye in the media⁴⁸[↗](#). Under aerobic conditions in rich media, we observed the expected rapid killing of *Mabs* over the first 5 days with the combination of TIG/LZD, with more rapid loss of viability in *KatG*⁻ cells. However, under hypoxic conditions, where ROS production is suppressed, we saw much slower bacterial killing. Importantly, under hypoxic conditions *katG*⁻ cells no longer had a survival defect relative to *katG*⁺ cells, supporting the hypothesis that translation-inhibiting antibiotics also cause secondary accumulation of lethal ROS in antibiotic-treated cells that need to be detoxified by KatG (**Figure 5F** [↗](#)).

We also evaluated whether other methods of alleviating ROS damage might enhance survival. The iron chelator 2,2'-dipyridyl has been shown in other contexts to reduce ROS-mediated damage by suppressing the reaction of H₂O₂ with Fe²⁺ that generates highly oxidizing hydroxyl radicals (Fenton reaction), and which has been shown to mitigate oxidative damage in other bacteria following antibiotic exposure²³[↗](#), ²⁶[↗](#), ⁴⁹[↗](#). As seen in other bacteria, we find that in *Mabs*, 2,2'-dipyridyl does indeed improve bacterial survival following exposure to bactericidal translation inhibitors, further supporting a role in ROS in cell death following translation inhibition (**Figure 5G** [↗](#)). We also assessed whether free radical scavengers like thiourea and 2,2,6,6-tetramethylpiperidine-1-oxyl (TEMPO) ameliorated antibiotic toxicity, although similar antioxidants had previously been shown in *Mtb* to increase respiration and paradoxically to increase ROS generation²⁷[↗](#). When we treated *Mabs* simultaneously with TIG/LZD in combination with either thiourea or TEMPO we did not observe a restoration of antibiotic tolerance, and, similar to observations in *Mtb*, actually observed increased bacterial cell death (**Figure 5 Supplement** [↗](#)).

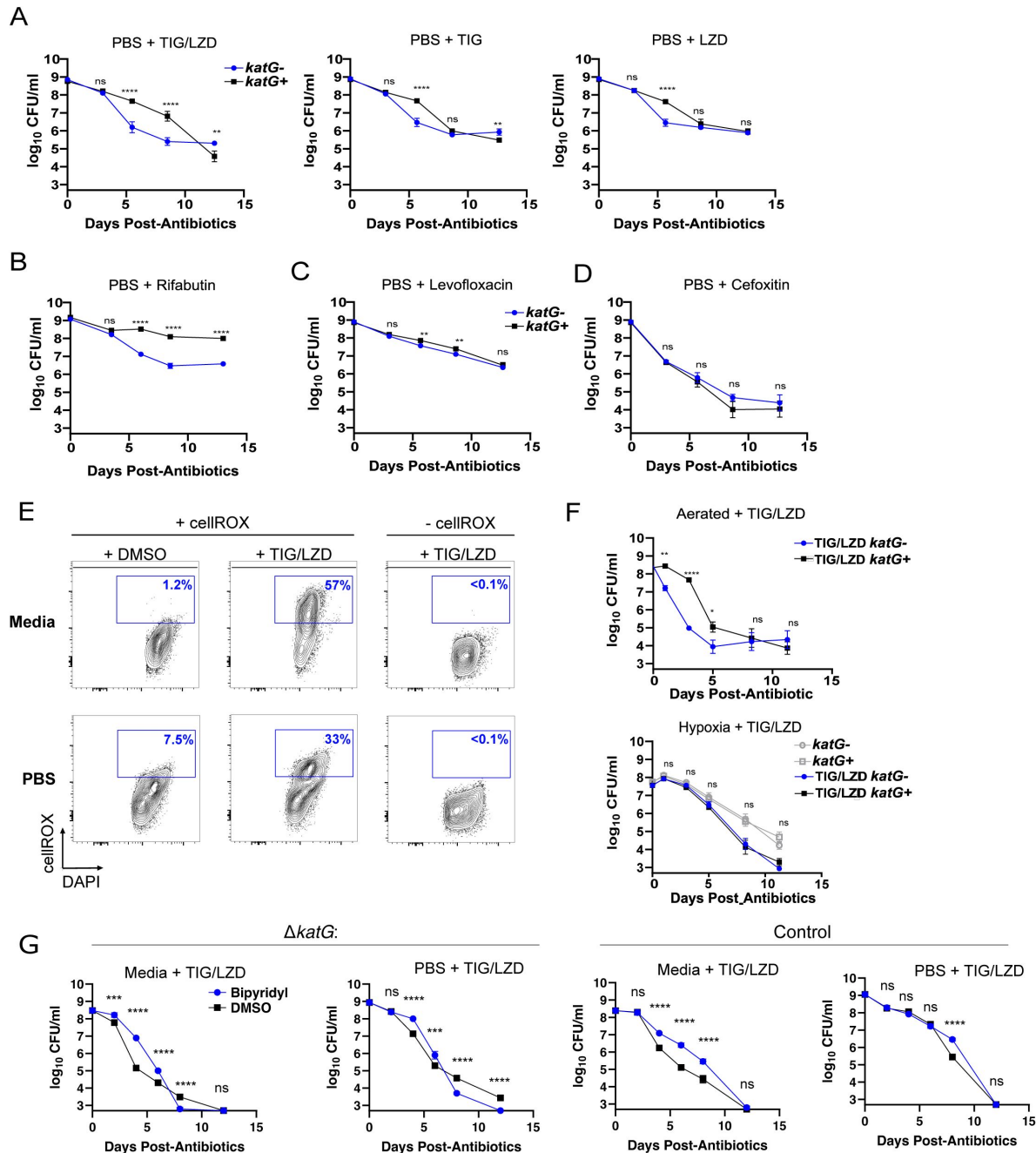


Figure 5.

ROS-mediated toxicity following antibiotic exposure.

(A-D) Analysis of *katG*⁺/*katG*⁻ cells challenged with different antibiotics. Cells were starved in PBS for 48h and then exposed to the indicated antibiotic. (E) Flow cytometry of control cells exposed to TIG/LZD (4h in 7H9 media or 72h in PBS) and then stained with DAPI and the ROS-sensitive dye cellROX green; percentage cellROX-positive cells are indicated. (F) Survival over time for aerated and hypoxic cultures of *Mabs* after exposure to TIG/LZD. (G) Survival over time for bipyridyl treated cells after exposure to TIG/LZD. Error bars represent SEM, statistical significance is calculated at each time point using student's t test. ****: $p < 0.0001$, ***: $p < 0.001$, **: $p < 0.01$, *: $p < 0.05$, ns: $p > 0.05$. (A-D, F) display combined data from 3 independent experiments. (E, G) are representative data from 3 independent experiments.

Incomplete penetrance of *katG* phenotype among *Mabs* strains

To test whether ROS accumulation was an effect occurring more broadly across different *Mabs* strains we obtained two clinical strains, exposed them to TIG/LZD in 7H9 media, and measured ROS accumulation with cellROX as above. We found that similar to ATCC 19977, both clinical *Mabs* strains had elevated ROS levels following translation inhibition (**Figure 6A**), suggesting that this is a conserved process in *Mabs*. Next, we tested the role of KatG in these *Mabs* clinical strains. We used ORBIT to disrupt the *katG* locus and evaluated the ability of these $\Delta katG$ clinical strains to survive exposure to TIG/LZD under both stressed and un-stressed conditions. Unlike ROS accumulation, where the responses across strains were consistent, we saw a variable dependency on *katG*. Clinical strain-1 behaved differently overall, with no appreciable starvation-induced antibiotic tolerance, and no contribution of *katG* to survival. In contrast, for clinical strain-2, *katG* contributed significantly to starvation-induced antibiotic tolerance, behaving similarly to the ATCC 19977 reference strain. However, unlike the reference strain, *katG* was not required for survival of clinical strain-2 when exposed to antibiotics in 7H9 media (**Figure 6B**). Thus, although antibiotic-induced ROS accumulation was observed across all three *Mabs* strains, the $\Delta katG$ phenotype displays incomplete penetrance, suggesting that in some *Mabs* strains alternative pathways exist that are able to compensate for the loss of *katG*.

In summary, the results of these studies point to an important effect of ROS in amplifying the lethality of transcription and translation-inhibiting antibiotics in *Mabs*. Through genetic analysis we identified a number of ROS detoxification factors, including KatG, as necessary for survival in this context. This suggested that antibiotics induced an oxidative state in cells, and direct measurement of ROS following antibiotic exposure indicated that this was indeed the case. Further supporting the toxic effects of ROS in this context, we found that removal of oxygen both slowed bacterial killing and rendered KatG dispensable. Taken together, these results suggest that in *Mabs* antibiotic lethality is accelerated by ROS accumulation, and that survival requires active detoxification systems.

Discussion

Pathways necessary for antibiotic tolerance in *Mabs*

The phenomenon of antibiotic tolerance has been recognized for decades and has been observed in a broad array of bacterial species, but without a uniform underlying mechanism - suggesting that different pathways may play roles in different physiologic contexts. For example, the pathways identified in one bacterial species may not contribute to tolerance in another. In addition to *E. coli*, *relA* plays a role in tolerance in *Pseudomonas aeruginosa*⁵³, *Staphylococcus aureus*⁵⁴, and *Mtb*⁵⁵. However, its role is not universal. Deletion of *relA* had no effect on antibiotic tolerance in *Msmeg*⁵⁶, and in our Tn-Seq analysis, *Mabs relA* Tn mutants had no survival defect. However, a prior study of the *Mabs relA* mutant demonstrated that this strain still synthesizes (p)ppGpp⁵⁷, suggesting potential genetic redundancy. Even within a single species there can be differences in the critical survival mechanisms depending on the context. In *E. coli*, RelA contributes to persister viability following exposure to β -lactams, but not quinolones⁵³, and in our study we find KatG to be essential for tolerance to transcription and translation inhibitors but not to β -lactams or quinolones.

Mechanisms of antibiotic lethality

Our findings strongly support the idea that ROS can be a significant contributor to antibiotic lethality in *Mabs* and contribute to the growing evidence that this is conserved across mycobacteria. In *Mabs*, *Mtb* and *Msmeg*, other groups have also observed reduced antibiotic-

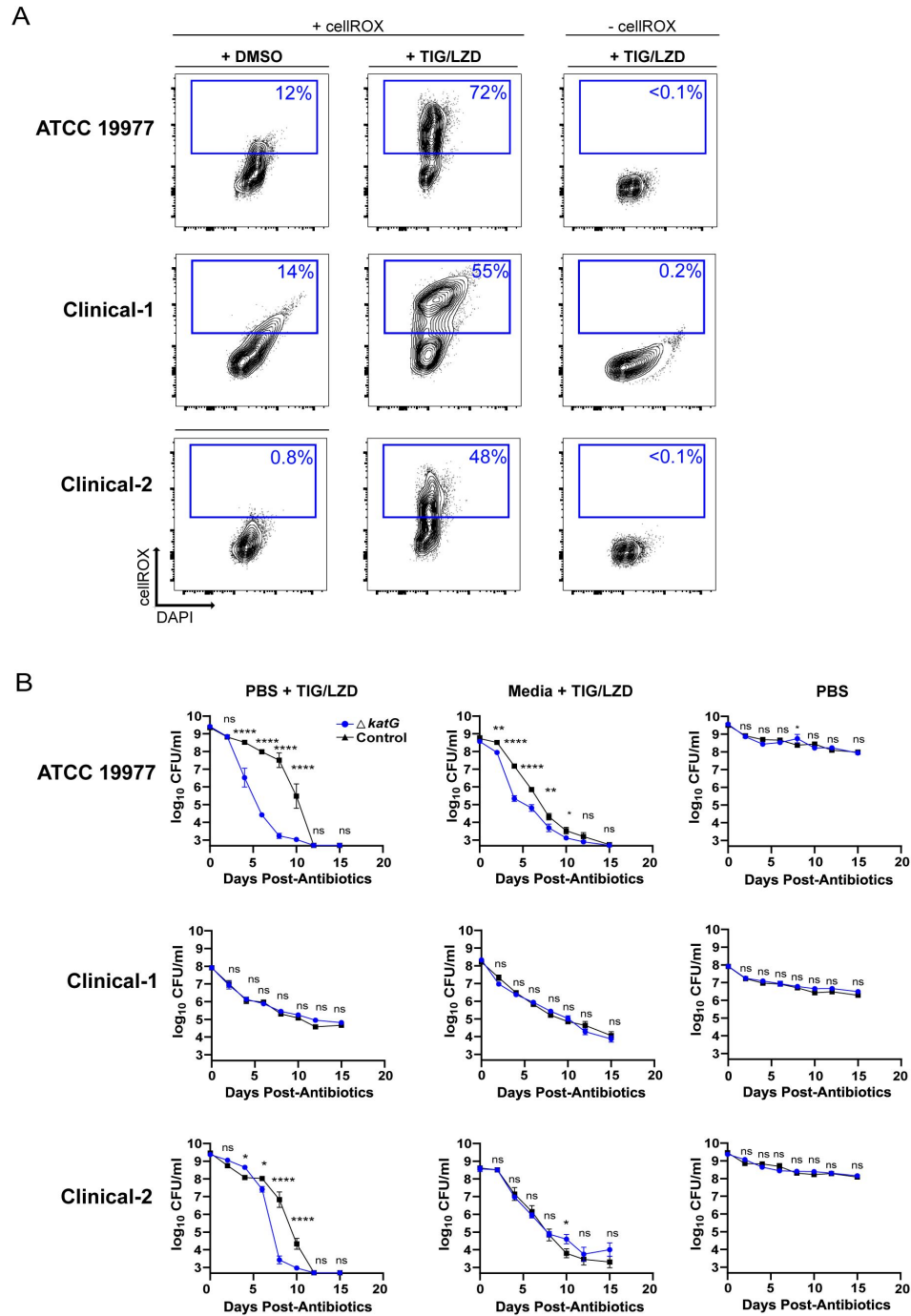


Figure 6.

Consistent antibiotic-induced ROS production but variable protection by KatG among different *Mabs* strains.

(A) The indicated strains of *Mabs* were cultured in 7H9 media, exposed to TIG/LZD for 4h and then analyzed by cellROX staining. (B) CFU over time following TIG/LZD exposure. Error bars represent SEM, statistical significance is calculated at each time point using student's t test. ****: $p < 0.0001$, ***: $p < 0.001$, **: $p < 0.01$, *: $p < 0.05$, ns: $p > 0.05$. Combined data from 4 independent experiments are shown for persister survival experiments. Representative data from 2 independent experiments are shown for flow cytometry experiments. Antibiotics were added as described above.

mediated killing in hypoxic conditions^{25, 37}. In addition, in *Mtb*, exposure to rifampin also generates ROS^{19, 60}, and *katG* contributes to survival in rifampin-treated cells²⁶.

Exactly how transcription or translation blockade leads to increased ROS levels is also not known. In principle, any of several derangements could lead to ROS accumulation. One of the major sources of cellular ROS is oxidative phosphorylation, as hydrogen peroxide and superoxide are natural byproducts. Thus, increased ROS generation by oxidative phosphorylation is an attractive hypothesis. Alternatively, particularly under starvation conditions, it is possible that antioxidants and ROS scavengers may become depleted, creating a more oxidizing environment. Our Tn-Seq analysis provides additional insight on this. We noted a small class of Tn mutants that were paradoxically protected from antibiotic lethality (**Figure 2B**). Prominent among this class of mutants were several independent components of the NADH dehydrogenase complex. Also known as Complex I of the electron transport chain, it is one of the key entry points for electrons into the oxidative phosphorylation pathway. The observation that mutants lacking this complex are protected suggests that decreasing flux through oxidative phosphorylation, with a concomitant decrease in ROS generation, may enhance survival during antibiotic exposure. A mechanistic understanding of how blockade of either transcription or translation leads to deranged oxygen utilization is an unresolved question that will require further study.

ROS accumulation is not universal following exposure to bactericidal antibiotics. While antibiotic-induced ROS is well-documented, under some conditions, such as higher concentrations of antibiotic, multiple studies have also observed antibiotic lethality without ROS accumulation^{23, 58–61}. Similarly, prior studies have found that under certain conditions, *E. coli* mutants lacking catalase have defects in antibiotic tolerance^{62–64}, whereas under other conditions they do not^{23, 65}. In *Mabs* the role of *katG* was also not uniform, as it had no impact on survival following exposure to levofloxacin or cefoxitin. Additional studies will be needed to determine whether levofloxacin and cefoxitin kill without generating ROS, or whether ROS is generated but effectively detoxified in the absence of *katG*. This latter possibility is suggested by our findings in *Mabs* clinical strains. While we saw ROS accumulation in both strains after TIG/LZD exposure, the role of *katG* was variable between the strains. This suggests that compensatory pathways likely exist and that in *Mabs* they can overcome the loss of *katG*.

Therapeutic implications

Mabs infections are particularly challenging to treat, with relapse rates over 50%³. Our results highlight several bacterial processes, such as the bacterial proteasome and ROS detoxification, which might be targeted therapeutically to reduce the survival of antibiotic tolerant bacteria in patients with *Mabs* infection. Agents targeting these processes might not have any intrinsic antimicrobial activity alone, but might act to disrupt the unique physiology required to survive antibiotics. This would represent a new therapeutic class of “persistence inhibitors” that might act synergistically with traditional antibiotics to eliminate the subpopulation of cells that would otherwise remain viable despite prolonged antibiotic treatment in patients with *Mabs* and other chronic infections^{6, 27}.

Limitations

Tn-Seq has inherent drawbacks, including an inability to identify mutants in essential genes, or in cases of genetic redundancy. Thus, there are likely genes needed for antibiotic tolerance in *Mabs* that were not identified in this study. In addition, we studied the response to a single class of antibiotic, focusing on the translation inhibitors often used to treat *Mabs* infections, and we studied only spontaneous persister cells and starvation-induced antibiotic tolerance. It is likely that examining other antibiotics, with different mechanisms of action, or different stresses that induce tolerance would identify additional genes contributing to survival and would allow identification of core pathways that might be shared in differing physiologic contexts of antibiotic and stress^{25, 27}.

Materials and methods

Key Reagent Table

Reagent	Source/Reference	Identifier
<i>Mabs</i> ATCC 19977	ATCC	19977
<i>Msmeg</i> MC2 155	ATCC	700084
<i>Mtb</i> Erdman	ATCC	35801
<i>Mabs</i> clinical strains	Sacramento Department of Public Health	NA
pkm444	Addgene	108319
pkm496	Addgene	109301
pmv306	Ref. ⁶⁶	
Middlebrook 7H9	BD	271310

Middlebrook 7H10	BD	262710
Glycerol	Invitrogen	15514011
Tween-80	Fisher	BP338-500
Middlebrook OADC	BD	212351
DPBS (-Ca/Mg)	Gibco	14190144
Tyloxapol	Sigma Aldrich	T8761
Methylene blue	Sigma Aldrich	28514
Tigecycline	Chem Impex	29737
Linezolid	Chem Impex	29723
Levofloxacin	Sigma Aldrich	28266
Cefoxitin	Chem Impex	1490
Rifabutin	Cayman Chemical	16468
Rifampin	Sigma Aldrich	R7382
Isoniazid	Supelco	I3377
Ethambutol	Thermo Scientific	J6069506
DMSO	Sigma Aldrich	D2650
Trizol	Invitrogen	15596018
0.1mm zirconia beads	Biospec	11079101z
PureLink RNA Mini Kit	Invitrogen	12183025
Kanamycin	VWR	75856-686
Zeocin	Invivogen	ANTZN1P
Glycine	Fisher Bioreagents	BP381
Sucrose	Sigma Aldrich	S1888
Anhydrotetracycline	Cayman Chemical	10009542
cellROX green	Invitrogen	C10444

Paraformaldehyde	Sigma Aldrich	P6148
DAPI	Sigma Aldrich	D9542
DNAseI	New England Biolabs	M0303
Maxima H minus reverse transcriptase	Thermo Scientific	EP0752
Taq Polymerase	New England Biolabs	M0273
EvaGreen	Biotium	31019
Bipyridyl	Sigma Aldrich	D216305
Thiourea	Sigma Aldrich	T7875
4-Hydroxy-TEMPO	Sigma Aldrich	176141

Bacterial strains and culture conditions

Mabs ATCC 19977, clinical *Mabs* strains and *Msmeg* (MC2 155) were grown in BD Middlebrook 7H9 media (liquid) or 7H10 media (solid) supplemented with 0.5% glycerol (Sigma) and 0.2% Tween-80 (Fisher) but without any OADC supplementation except for transformations. Sacramento clinical isolates were obtained from the Sacramento County Department of Public Health Mycobacteriology Laboratory. Confirmation of clinical isolates as *Mabs* was performed by amplifying the 16s rRNA locus and Sanger sequencing (maybe reference new data table?). *Mtb* (Erdman) was grown in 7H9 (liquid) or 7H10 (solid) supplemented with 0.5% glycerol, 0.1% Tween-80, and 10% OADC (BD). All cultures were grown at 37°C with gentle shaking. Except for specific hypoxia conditions, all liquid cultures were grown with 90% container headspace or using a gas permeable cap to ensure culture oxygenation. PBS starvation was achieved by washing OD0.5-1.0 *Mabs* 1X in DPBS (-Ca/Mg, Gibco) and resuspending in DPBS at OD=1, supplemented with 0.1% tyloxapol (Sigma).

Mabs antibiotic experiments

For PBS starvation experiments, stocks of *Mabs* were grown for 48 hours in 7H9 passaging continuously in log phase, then PBS starved or passaged in log phase for an additional 48h. Log phase or PBS starved *Mabs* were then resuspended in antibiotic containing media at OD1.0. For experiments with hypoxia, *Mabs* in mid-log aerobic growth was adjusted to OD0.001 in media with 1.5µg/ml methylene blue and added to a rubber septum sealed glass vial with 50% headspace. Methylene blue discoloration was observed at 3d and antibiotics were added at 5d. For thiourea and TEMPO experiments, these were added to the cultures at the time of antibiotic administration. For bipyridyl experiments, 62.5µM bipyridyl was added to the cultures 2h prior to antibiotic administration. We empirically determined the half-life of each antibiotic in 7H9 media at 37-deg and for those with half-lives shorter than the experiment, supplemented cultures to with additional antibiotic to maintain the concentration of active antibiotic. Antibiotics were used at the following concentrations: tigecycline (Chem-Impex) at 10µg/ml (8 fold above MIC, re-administered every 3 days), linezolid (Chem-Impex) at 100µg/ml (20 fold above MIC), levofloxacin (Sigma) at 40µg/ml (8 fold above MIC), cefoxitin (Chem Impex) at 80µg/ml (8 fold above MIC, re-administered every 3 days), and rifabutin (Cayman) at 40µg/ml (4 fold above MIC). After antibiotic

administration, colony forming units over time were measured by plating on 7H10 solid media. For experiments where growth recovery time in liquid media was quantified, 100µl of sample was removed at d6 after antibiotic administration and washed 2X in antibiotic-free media. The samples were resuspended in 5ml of antibiotic free media, and OD620 measurements were taken with a FilterMax F3 plate reader (Molecular Devices) until maximum cell density (OD of ~5.0) was reached.

***Msmeg* antibiotic experiments**

Individual colonies were picked and grown for 48 hours in log phase before being PBS starved or passaged in log phase for 48h. Log phase or PBS starved *Msmeg* were then resuspended in antibiotic containing media at OD1.0. Antibiotics were used at the following concentrations: tigecycline (Chem-Impex) at 1.25µg/ml (8 fold above MIC, re-administered every 3 days), linezolid (Chem-Impex) at 2.5µg/ml (8 fold above MIC), rifampin (Sigma) at 32µg/ml (8 fold above MIC, re-administered every 6 days), isoniazid (Sigma) at 32µg/ml (8 fold above MIC, re-administered every 6 days), and ethambutol (Thermo) at 4µg/ml (8 fold above MIC, re-administered every 3 days). After antibiotic administration, colony forming units over time were measured.

***Mtb* antibiotic experiments**

Freezer stocks of *Mtb* were thawed and grown for 5-7 days in log phase before being starved for 14d or longer. Non-starved control *Mtb* were thawed such that they were also grown for 5-7 days in log phase before experimental use. Log phase or PBS starved *Mtb* was then resuspended in antibiotic containing media and adjusted to OD1.0. Antibiotics were used at the following concentrations: rifampin (Sigma) at 0.1µg/ml (4 fold above MIC, re-administered every 6 days), isoniazid (Sigma) at 0.1µg/ml (4 fold above MIC, re-administered every 6 days), and ethambutol (Thermo) at 8µg/ml (4 fold above MIC, re-administered every 6 days). After antibiotic administration, colony forming units over time were measured.

Transposon insertion sequencing

The construction of this *Himar1* transposon Tn library has been described previously³³. Screening was performed by growing a freezer stock of the library for 2.5 days in log phase before 48-hour PBS starvation or further continuous log-phase growth. Samples were then resuspended in media containing either tigecycline/linezolid or an equal volume of DMSO solvent and incubated for 6 days, with a re-administration of tigecycline or matching DMSO on day 3. Cultures were aerated by culturing in a vented cap bottle with gentle agitation at 40 revolutions per minute throughout the experiment. The samples were then washed 2X in antibiotic free liquid media, resuspended in antibiotic free liquid media (10X the original culture volume), and grown until OD0.5-1.0.

Subsequently, the samples underwent three more rounds of 100-fold passaging in liquid media to amplify surviving bacteria before the samples were collected in Trizol (Invitrogen). A sample taken at the time of the commencement of PBS starvation was collected in Trizol and used as the input control. Three independent trials of this experiment were submitted to the UC Davis DNA Technologies Core, where Tn insertion site flanking sequences were amplified as described previously³³ and sequenced on an Element Biosciences AVITI. Sequence reads were mapped to the ATCC 19977 genome and analyzed using TRANSIT software with the following parameters: 0% of N/C termini ignored, 10,000 samples, TTR normalization, LOESS correction, include sites with all zeros, site restricted resampling. Genes with significant changes were defined as those with adjusted p-value (p-adj.) <0.05 and log₂ fold change >0.5. P-adj. was calculated using the Benjamini-Hochberg correction.

Pathway enrichment analysis

To improve gene annotation, *Mabs* orthologs to *Mtb* genes were identified. *Mabs* genes were first converted into protein sequences using Mycobrowser, and protein sequences were then used to perform reciprocal BLASTp searches. *Mabs* genes and *Mtb* genes that mapped to each other using independent one-way BLASTp searches with a maximum e-value cutoff of 0.1 were considered orthologs. For pathway analysis, gene lists (*Mtb* orthologs) were then imported into the DAVID knowledgebase⁴² and pathway enrichment analysis performed for Gene Ontology biological process, Uniprot keyword and KEGG databases with statistical analysis Fisher's exact test and nominal p-value reported.

Gene deletion and complementation

Knockout strains were generated using ORBITT⁴⁸. Briefly, *Mabs* was transformed with the kanamycin-resistant ORBIT recombineering plasmid pkm444. 20ml *Mabs* at OD0.5-1.0 was washed 2X in 10% glycerol and resuspended in 200ul 10% glycerol. 500ng plasmid was added and electroporated at 2.5kV in 0.2cm cuvettes. The bacteria were allowed to recover overnight before plating on 150µg/ml kanamycin plates. Clones were selected and regrown in liquid media supplemented with 150µg/ml kanamycin and 10% OADC (BD) to OD0.5-1.0. For recombineering, the pkm444-*Mabs* was grown to mid-log, then diluted to OD0.1 and 200mM glycine (Fisher) was added to the media. 16 hours later, 500mM sucrose (Sigma) and 500ng/ml anhydrotetracycline (Cayman) were added and incubated for an additional 4 hours. Subsequently, the *Mabs* was washed 2X in ice cold 10% glycerol + 500mM sucrose. 200ul of 10X concentrated *Mabs* was then electroporated with 600ng of the zeocin-R ORBIT payload pkm496 plasmid and 2µg of targeting oligonucleotide (Table S3) at 2.5kV in 0.2cm cuvettes. The *Mabs* was then allowed to recover overnight in liquid media with 10% OADC and 500ng/ml anhydrotetracycline before being plated on 150µg/ml zeocin plates. Mutants were then selected and screened for gene deletion by PCR amplification and Sanger sequencing. For genetic complementation, the endogenous loci including promoter and terminator sequences were amplified by PCR and cloned into the EcoRV site of pmv306 with kanamycin resistance⁶⁶. In the case of *katG*, the upstream gene *furA* was also included in the complementation construct to achieve optimal *katG* expression.

MIC determination

Two-fold serial dilutions of antibiotics were prepared in a 96 well plate in 100ul volume. 100ul of 2X bacteria were added (for *Mabs*: used a final OD of 0.001, *Msmeg*: OD0.001, *Mtb*: OD0.01), making a final volume of 200µl. The plates were incubated until there was visible growth in the no antibiotic control well. At this time, the bacteria were transferred to a new plate with 20µl of 40% paraformaldehyde and OD620 measurements were taken with a FilterMax F3 plate reader (Molecular Devices).

Flow cytometry

OD1 Mab was stained with cellROX green (Invitrogen) at a final concentration of 5µM for 1hr at 37C. The cells were then washed in PBS and resuspended in PBS with 4% paraformaldehyde and 5µg/ml DAPI (Sigma). The samples were run on an LSRII flow cytometer (BD). Fluorophores were excited with the 405nm (DAPI) and 488nm (cellROX) lasers. Detection was performed using the 450/50 (505LP) filter for DAPI and a 525/50 (555LP) filter for cellROX. Data were analyzed with FlowJo software (BD).

DNA/RNA Purification

Samples were resuspended in 5 volumes of Trizol, and bead beat with 0.1mm zirconia beads (Biospec) 6x2min at 4°C in a Mini-Beadbeater-16 (Biospec). Chloroform was added and RNA in the aqueous phase removed. For DNA isolation, a second RNA extraction was performed with 0.8M guanidine thiocyanate and 0.5M guanidine hydrochloride, 60mM Acetate pH 5.2, 1mM EDTA. DNA was then isolated with back-extraction buffer (4M Guanidine Thiocyanate, 50mM Sodium Citrate, 1M Tris base (without pH adjustment ~pH 11) and DNA purified using a PureLink RNA Mini Kit (Invitrogen).

RT-qPCR

RNA was purified using PureLink RNA Mini Kit per manufacturer's instructions. The samples were DNaseI (NEB) treated for 15min/37°C before stopping the reaction by adding 3.5mM EDTA and heating for 10min/75°C. cDNA was synthesized from 500ng total RNA using random hexamers and Maxima H minus reverse transcriptase (Thermo). No reverse-transcription controls were also included and used to confirm the lack of genomic DNA-driven amplification. qPCR reactions used Taq polymerase (NEB) and EvaGreen (Biotium) and were run on Biorad CFX Opus 96 Real-Time PCR System. Melt curves were included for each sample to confirm uniform amplicon identity between samples. Gene-specific amplification was quantified by comparison to a standard curve generated from 3-fold serial dilutions of a control sample, then normalized to 16S rRNA within each sample.

Figure 3 Supplement.

Additional analysis of mutants.

(A) *blaR* or (B) *recR* mutants were examined under conditions where Tn-Seq did not predict a phenotype. Cells were starved in PBS for 48h prior to treatment with TIG/LZD or DMSO as described in [Figure 3](#). (C) Growth recovery of mutants after antibiotic exposure. After 6 d of TIG/LZD exposure cells were washed twice in antibiotic free media and inoculated into 7H9 media. Growth was monitored by OD₆₀₀ of the culture. Error bars represent SEM, statistical significance is calculated at each time point using student's t test. ****: $p < 0.0001$, ***: $p < 0.001$, **: $p < 0.01$, *: $p < 0.05$, ns: $p > 0.05$. (A-B) are representative data from 4 independent experiments (C) is representative data from 2 independent experiments.

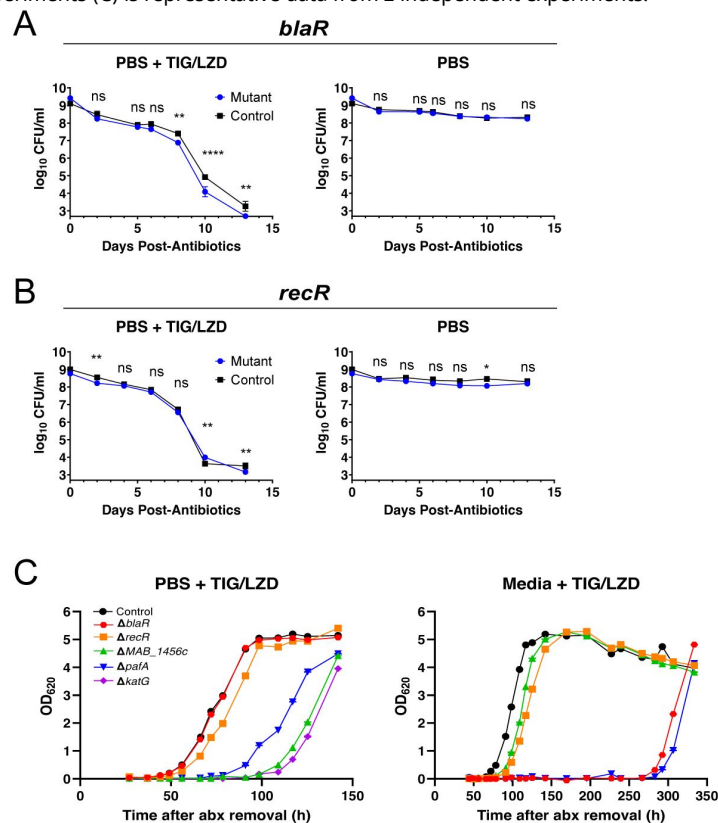
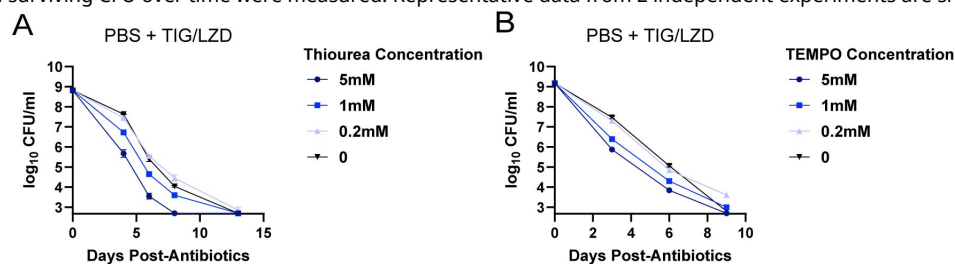


Figure 5 Supplement.

Effect of ROS scavengers.

ΔkatG cells were starved in PBS for 48h then treated with TIG/LZD and the indicated concentration of (A) thiourea or (B) TEMPO and surviving CFU over time were measured. Representative data from 2 independent experiments are shown.



Data availability

Numerical source data for all figures has been deposited at Dryad <https://doi.org/10.5061/dryad.4xgxd25pv>

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Additional files

Supplemental Table 1. *Tn-Seq results.*

Supplemental Table 2. *DAVID functional pathway analysis.*

Supplemental Table 3. *Oligonucleotide sequences.*

Supplemental Table 4. *16s sequencing data of clinical strains.*

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Reviewer #1 (Public review):

Summary:

Persistence is a phenomenon by which genetically susceptible cells are able to survive exposure to high concentrations of antibiotics. This is especially a major problem when treating infections caused by slow growing mycobacteria such as *M. tuberculosis* and *M. abscessus*. Studies on the mechanisms adopted by the persisting bacteria to survive and evade antibiotic killing can potentially lead to faster and more effective treatment strategies.

To address this, in this study, the authors have used a transposon mutagenesis based sequencing approach to identify the genetic determinants of antibiotic persistence in *M. abscessus*. To enrich for persisters they employed conditions, that have been reported previously to increase persister frequency - nutrient starvation, to facilitate genetic screening for this phenotype. *M. abscessus* transposon library was grown in nutrient rich or nutrient depleted conditions and exposed to TIG/LZD for 6 days, following which Tn-seq was carried out to identify genes involved in spontaneous (nutrient rich) or starvation-induced conditions.

About 60% of the persistence hits were required in both the conditions. Pathway analysis revealed enrichment for genes involved in detoxification of nitrosative, oxidative, DNA damage and proteostasis stress. The authors then decided to validate the findings by constructing deletions of 5 different targets (pafA, katG, recR, blaR, Mab_1456c) and tested the persistence phenotype of these strains. Rather surprisingly only 2 of the 5 hits (katG and pafA) exhibited a significant persistence defect when compared to wild type upon exposure to TIG/LZD and this was complemented using an integrative construct. The authors then investigated the specificity of delta-katG susceptibility against different antibiotic classes and demonstrated increased killing by rifabutin. The katG phenotype was shown to be mediated through the production of oxidative stress which was reverted when the bacterial cells were cultured under hypoxic conditions. Interestingly, when testing the role of katG in other clinical strains of Mab, the phenotype was observed only in one of the clinical strains demonstrating that there might be alternative anti-oxidative stress defense mechanisms operating in some clinical strains.

Strengths:

While the role of ROS in antibiotic mediated killing of mycobacterial cells have been studied to some extent, this paper presents some new findings with regards to genetic analysis of *M. abscessus* susceptibility, especially against clinically used antibiotics, which makes it useful. Also, the attempts to validate their observations in clinical isolates is appreciated.

Weaknesses:

Amongst the 5 shortlisted candidates from the screen, only 2 showed marginal phenotypes which limits the impact of the screening approach.

While the role of KatG mediated detoxification of ROS and involvement of ROS in antibiotic killing was well demonstrated, the lack of replication of this phenotype in some of the clinical isolates limits the significance of these findings.

<https://doi.org/10.7554/eLife.104944.2.sa3>

Reviewer #2 (Public review):

Summary:

The work set out to better understand the phenomenon of antibiotic persistence in mycobacteria. Three new observations are made using the pathogenic *Mycobacterium abscessus* as an experimental system: phenotypic tolerance involves suppression of ROS, protein synthesis inhibitors can be lethal for this bacterium, and levofloxacin lethality is unaffected by deletion of catalase, suggesting that this quinolone does not kill via ROS.

Strengths:

The ROS experiments are supported in three ways: measurement of ROS by a fluorescent probe, deletion of catalase increases lethality of selected antibiotics, and a hypoxia model suppresses antibiotic lethality. A variety of antibiotics are examined, and transposon mutagenesis identifies several genes involved in phenotypic tolerance, including one that encodes catalase. The methods are adequate for making these statements.

Weaknesses:

The work can be improved by a more comprehensive treatment of prior work, especially comparison of *E. coli* work with mycobacterial studies.

Moreover, the work still has some technical issues to fix regarding description of the methods, supplementary material, and reference formatting.

Overall impact: Showing that ROS accumulation is suppressed during phenotypic tolerance, while expected, adds to the examples of the protective effects of low ROS levels. Moreover, the work, along with a few others, extends the idea of antibiotic involvement with ROS to mycobacteria. These are field-solidifying observations.

Comments on revisions:

The authors have moved this paper along nicely. I have a few general thoughts.

(1) It would be helpful to have more references to specific figures and panels listed in the text to make reading easier.

(2) I would suggest adding a statement about the importance of the work. From my perspective, the work shows the general nature of many statements derived from work with *E. coli*. This is important. The abstract says this overall, but a final sentence in the abstract would make it clear to all readers.

(3) The paper describes properties that may be peculiar to mycobacteria. If the authors agree, I would suggest some stress on the differences from *E. coli*. Also, I would place more stress on novel findings. This might be done in a section called Concluding Remarks. The paper by Shee 2022 AAC could be helpful in phrasing general properties.

(4) Several aspects still need work to be of publication quality. Examples are the materials table and the presentation of supplementary material. Reference formatting also needs attention.

<https://doi.org/10.7554/eLife.104944.2.sa2>

Reviewer #3 (Public review):

Summary:

The manuscript demonstrates that starvation induces persister formation in *M. abscesses*. They also utilized Tn-Seq for the identification of genes involved in persistence. They identified the role of catalase-peroxidase KatG in preventing death from translation inhibitors Tigecycline and Linezolid. They further demonstrated that a combination of these translation inhibitors leads to the generation of ROS in PBS-starved cells.

Strengths:

The authors used high-throughput genomics-based methods for identification of genes playing a role in persistence.

Weaknesses:

The findings could not be validated in clinical strains.

Comments on revisions: No more comments for the authors.

<https://doi.org/10.7554/eLife.104944.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Weaknesses:

Only 1 gene (katG) gave a strong and 1 (Mab_1456c) exhibited a minor defect. Two of the clones did not show any persistence phenotype (blaR and recR) and one (pafA) showed a minor phenotype,

We have now carried out more detailed validation studies on the Tn-Seq, with analysis of time-dependent killing over 14 d. This more comprehensive analysis shows that 4 of 5 genes analyzed do indeed have antibiotic tolerance defects under the conditions that Tn-Seq predicted a survival defect (Revised Figure 3). In addition, we found that even before actual cell death, several mutants had delayed resumption of growth after antibiotic removal (Figure 3 Supplemental).

Fig 3 - Why is there such a huge difference in the extent of killing of the control strain in media, when exposed to TIG/LZD, when compared to Fig. 1C and Fig. 4. In Fig. 1C, M. abs grown in media decreases by >1 log by Day 3 and >4 log by Day 6, whereas in Fig. 3, the bacterial load decreases by <1 log by Day 3 and <2 log by Day 6. This needs to be clarified, if the experimental conditions were different, because if comparing to Fig. 1C data then the katG mutant strain phenotype is not very different.

We agree with the reviewer that there is variability in the timing and extent of cell death from experiment to experiment. As noted by the reviewer, in Figure 1C the largest decrement in survival is between day 1 - day 3 (also seen in Figure 6A). As they noted in Figure 4 the largest decrement is between day 3 - day 6 (also seen in Figure 3A, Figure 5F). In each experiment with katG mutants we carefully compare the mutant vs. the control strain within that experiment, which is more accurate than comparing the behavior of mutant in one experiment to a control in another experiment.

Reviewer #2 (Public review):

Weaknesses:

.First, word-choice decisions could better conform to the published literature. Alternatively, novel definitions could be included. In particular, the data support the concept of phenotypic tolerance, not persistence.

We appreciate the reviewers comments, text modified.

Second, two of the novel observations could be explored more extensively to provide mechanistic explanations for the phenomena.

We have added several additional experiments, these are detailed below in response to specific comments.

Reviewer #3 (Public review):

Weaknesses:

The findings could not be validated in clinical strains.

We understand the reviewer's concern that the katG phenotype was only observed in one of the two clinical strains we studied. We feel that our findings are relevant beyond the ATCC 19977 strain for two reasons

(1) We have performed additional analyses of the two clinical isolates and indeed find significant accumulation of ROS following antibiotic exposure in both of these strains (revised Figure 6A).

(2) We do in fact see a role for katG in starvation-induced antibiotic tolerance in Mabs clinical strain-2. It is not surprising that different strains from a particular species may have some different responses to stresses – for example, there is wide strain-specific variability in susceptibility to different phages within a species based on which particular phage defense modules a given strain carries (for example PMID: 37160116). We speculate that different Mabs strains may express varying levels of other antioxidant factors and note that the genes encoding several such factors were identified by our Tn-Seq screen including the peroxidases ahpC, ahpD, and ahpE. Our analysis of the genetic interactions between katG and these other factors is ongoing.

Comments/Suggestions

(1) In Fig1E, the authors show no difference in killing Mtb with or without adaptation in PBS. These data are contrary to the data presented in Figure 1B. These also do not align with the data of M. smegmatis and M. abscesses. Please discuss these observations in light of the Duncan model of persistence (Mol Microbiol. 2002 Feb;43(3):717-31.).'

The above referenced Duncan laboratory study found tolerance after prolonged starvation but did not actually examine tolerance at early time points. While some of the transcriptional and metabolic changes seen by Duncan and others are slow, other groups have described starvation responses in Mtb that are quite rapid. For example, the stringent response mediator ppGpp accumulates within a few hours after onset of starvation in Mtb (PMID: 30906866). We suspect that a rapid signaling response such as this underlies the phenotype we observe. Regarding the difference between Mtb and other mycobacterial species we also find it surprising that Mtb had a much more rapid starvation response. This is a clear species-specific difference that may reflect an adaptation of Mtb to the nutrient-limited physiologic niche within host macrophages.

(2) Line 151, the authors state that they have used an M. abscesses Tn mutant library of ~55,000 mutant strains. The manuscript will benefit from the description of the coverage of total TA sites covered by the mutants.

Text modified to add this detail. There are 91,559 TA sites in the abscessus genome. Thus, our Tn density is ~60%.

(3) Line 155: Please explain how long the cells were kept in an Antibiotic medium.

This technical detail was noted above on line 153 in the original text: "...and then exposed them to TIG/LZD for 6 days". To clarify the overall conditions, we have also revised the text of the manuscript and added the detail of how long cells were passaged after removal of antibiotics.

(4) Line 201: data not shown. Delayed resumption of growth after removal of antibiotic would be helpful in indicating drug resilience. This data could enhance the manuscript.

Data now provided in Figure 3 Supplemental

(5) Figures 4C and 4F represent the kill curve. It will be good to show the data with CFU against the drug concentration in place of OD600. CFU rather than OD600 best reflects growth inhibition.

Figures 4C and 4F are measuring the minimum inhibitory concentration (MIC) to stop the overall growth of the bacterial population. While we agree that CFU could be analyzed, this would be measuring a different outcome – cell death and the minimum bactericidal concentration (MBC). In these experiments we sought to specifically examine the MIC so as to separate growth inhibition from cell death. For this we used the standard method employed by clinical microbiology laboratories for MIC, which is optical density of the culture (PMID: 10325306).

(6) Figure 5C. The authors shall show the effect of TIG/LZD on M. abscesses ROS production without the PBS adaptation. It is important to conclude that TIG/LZD induces ROS in cells. Authors should utilize ROS scavengers such as Thiourea, DFO, etc., to conclude ROS's contribution to bacterial killing following inhibition of transcription and translation.

New data added (revised Figure 5 and Figure 5 Supplemental)

(7) Line 303. Remove "note".

Text revised. We thank the reviewer for identifying this typographical error.

(8) The introduction and Discussion are very similar, and several lines are repeated.

Text revised with overlapping content removed.

Reviewer #1 (Recommendations for the authors):

It appears that the same datasets for PBS adapted cultures were plotted in A-C and D-F. Either this should be specifically mentioned in the legend or it might be better to integrate the non-adapted plots into A-C which would also allow easier comparison.

Appreciate the reviewer's suggestion; text modified with added clarification to figure legend.

This manuscript is focused on M. abs and the antibiotics TIG/LZD, so the Mtb data or data using the antibiotics INH/RIF/EMB and serves more as a distraction and can be removed

We appreciate the reviewer's perspective. However, we wish to include these data to show the similarities (and differences) in starvation-induced tolerance between the three organisms.

Fig 3 -As mentioned for Fig. 1, it appears that the same dataset was used for the control in all the figures A-E. This should be explicitly stated in the Figure legend.

Appreciate the reviewer's suggestion; text modified with added clarification to figure legend.

The divergent results from the clinical strains are extremely interesting. It would be helpful to determine the oxidative stress levels (similar to the cellROX data shown in 5E), to tease out if the difference in katG role is because of lack of ROS induction in these strains or due to expression of alternate anti-oxidative stress defense mechanisms.

We have performed additional cellROX analysis as suggested by the reviewer and found that the ROS induction is indeed present across all three Mabs strains, but that katG is only required in one of the two strains (Strain #2). These data are now included in the revised Figure 6.

Reviewer #2 (Recommendations for the authors):

GENERAL COMMENTS

This is a nice piece of work that uses the pathogen Mabs as a test subject.

The work has findings that likely apply generally to antibiotics and mycobacteria: 1) phenotypic tolerance is associated with suppression of ROS, 2) lethal protein synthesis inhibitors act via accumulation of ROS, and 3) levofloxacin behaves in an unexpected way. Each is a new observation. However, I believe that each topic requires more work to be firmly established to be suitable for eLife.

Phenotypic tolerance: Association with suppression of ROS is important but expected. I would solidify the conclusion by performing several additional experiments. For example, confirm the lethal effect of ROS by reducing it with an iron chelator and a radical scavenger. There is a large literature on effects of iron uptake, levels, etc. on antibiotic lethality that could be applied to this question. In 2013 Imlay argued against the validity of fluorescent probes. Perhaps getting the same results with another probe would strengthen the conclusion.

We have carried out additional experiments with both an iron chelator and small molecule ROS scavengers to further test this idea but note that these experiments have several inherent limitations: 1) These compounds have highly pleiotropic effects. For example while N-acetyl cysteine (NAC) is an antioxidant it also increases mycobacterial respiration and was shown to paradoxically decrease antibiotic tolerance in *M. tuberculosis* (PMID: 28396391). 2) It has been shown by the Imlay group that small-molecule antioxidants are often ineffective in quenching ROS in bacteria (PMID: 388893820), making negative results difficult to interpret. Nonetheless, we present new experimental data showing that iron chelation does indeed improve the survival of antibiotic-treated Mabs (revised Figure 5). However, small molecule antioxidants such as thiourea do not restore antibiotic tolerance and actually increased bacterial cell death, suggesting that they may be affecting respiration in Mabs in a manner similar to that seen for NAC in *Mtb*. We also note that our genetic analysis, which identified numerous other genes encoding proteins with antioxidant function (Figure 2) is a strong additional argument in support of the importance of ROS in antibiotic-mediated lethality.

Regarding the concern raised by Imlay about the validity of oxidation-sensitive dyes - this relates to concern bacterial autofluorescence induced by antibiotics that can confound analyses in some species. We have ruled this out in our analyses by using bacteria unstained by cellROX as controls to confirm that there is negligible autofluorescence in Mabs (<0.1%, Figure 5E, Figure 6A).

*Protein synthesis inhibitors: At present, this is simply an observation. More work is needed to suggest a mechanism. For example, with *E. coli* the aminoglycosides are protein synthesis inhibitors that also cause membrane damage. Membrane damage is known to stimulate ROS-mediated killing. Your observation needs to be extended because chloramphenicol, another protein synthesis inhibitor, blocks ROS production. The lethality may be a property of mycobacteria: does it occur with *E. coli* (note that rifampicin is bacteriostatic with *E. coli* but lethal to *Mtb*)?*

We agree with the reviewer that the mechanism underlying ROS accumulation following transcription or translational inhibition in Mabs is of significant interest. It is likely to be a mechanism different from *E. coli*, because in *E. coli* tetracyclines and rifamycins are both bacteriostatic, whereas in Mabs they are both bactericidal. Determining the mechanism by

which translation inhibitors cause ROS accumulation in Mabs is an ongoing effort in our laboratory using proteomics and metabolomics, but is outside the scope of this manuscript.

Levofloxacin: This is also at the observational stage but is unexpected. In other studies, ROS is involved in quinolone-mediated killing of bacteria. Why is this not the case with Mabs? The observation should be solidified by showing the contrast with moxifloxacin, since this compound has been studied with mycobacteria (Shee 2022 AAC). With E. coli, quinolone structure can affect the relative contribution of ROS to killing (Malik 2007 AAC), as is also seen with Mtb (Malik 2006 AAC). What is happening in the present work with levofloxacin, an important anti-tuberculosis drug? Is there a structure explanation (compare with ofloxacin)?

While these are interesting questions, a detailed exploration of the structure-function relationships between different fluoroquinolone antibiotics and their varying activities on Mtb and Mabs is outside the scope of this manuscript.

The writing is generally easy to follow. However, the concept of persistence should be changed to phenotypic tolerance with text changes throughout. I base this suggestion on the definitions of tolerance and persistence as stated in the consensus review (Balaban 2019 Nat Micro Rev). Experimentally, tolerance is seen as a gradual decline in survival following antibiotic addition; the decline is slower than seen with wild-type cells. The data presented in this paper fit that definition. In contrast, persistence refers to a rapid drop in survival followed by a distinct plateau (Balaban 2019 Nat Micro Rev; for example, see Wu Lewis AAC 2012). Moreover, to claim persistence, it would be necessary to demonstrate subpopulation status, which is not done. The Balaban review is an attempt to bring order to the field with respect to persistence and tolerance, since the two are commonly used without regard for a consistent definition.

We appreciate the reviewer's suggestion; text modified in multiple places to clarify.

Another issue requiring clarification is the relationship between resistance and tolerance. Killing by antibiotics is a two-step process, as most clearly seen with quinolones. First a reversible bacteriostatic event occurs. Resistance blocks that bacteriostatic damage. Then a lethal metabolic response to that damage occurs. Tolerance selectively blocks the second, killing event, a distinct process that often involves the accumulation of ROS. Direct antibiotic-mediated damage is an additional mode of killing that also stems from the reversible, bacteriostatic damage created by antibiotics. The authors recognize the distinction but could make it clearer. Take a look at Zheng (JJ Collins) 2020, 2022.

Text modified to clarify this point

Many readers would also like to see a bit more background on Mabs. For example, does it grow rapidly? Are there features that make it a good model for studying mycobacteria or bacteria in general? The more general, the better.

Text modified, background added

Below I have listed specific comments that I hope are useful in bringing the work to publication and making it highly cited.

SPECIFIC COMMENTS

Line 30 unexpectedly. I would delete this word because the result is expected from the ROS work of Shee et al 2022 with mycobacteria. Moreover, Zeng et al 2022 PNAS showed

that ROS participates in antimicrobial tolerance, and persistence is a form of tolerance (Balaban et al, 2019, Nat Micro Rev).

Text modified as per review suggestion

Line 39 key goal: this is probably untrue in the general sense stated, since bacteriostatic antibiotics are sufficient to clear infection (Wald-Dickler 2019 Clin Infect Dis). However, it is likely to be the goal for Mtb infections.

We agree with the reviewer that bacteriostatic antibiotics are effective in treating most types of infections and do not claim otherwise in the manuscript. However, from a clinical standpoint, eradication of the pathogen causing the infection is indeed the goal of antibiotic therapy in virtually all circumstances (with the exception of specific scenarios such as cystic fibrosis where it is recognized that the infecting organism cannot be fully eliminated). In most cases, the combination of bacteriostatic antibiotics and the host immune response is sufficient to achieve eradication. We have modified the manuscript text to reflect this nuance noted by the reviewer.

Line 62 several: you list three, but hipAB works via ppGpp, so the sentence needs fixing

Text modified

Line 70 uncertain: this uncertainty is unreferenced. Since everything is uncertain, this vague phrase does not add to the story.

The reviewer makes an interesting philosophical argument. However, we would submit that some aspects of biology, for example the regulation of glycolysis, are understood in great detail. However, other mechanisms, such as the precise mechanisms of lethality for diverse antibiotics in different bacterial species, are far more uncertain and remain a subject of debate (for example PMID: 39910302). Text not modified.

Line 72 somewhat controversial: I would delete this, because the points in the Science papers by Lewis and Imlay have been clarified and in some cases refuted by prior and subsequent work.

Text modified

Line 72 presumed: this suggests that it is wrong and perhaps a different idea has replaced it. Another, and more likely view is that there is an additional mode of killing. I suggest rephrasing to be more in line with the literature.

Text modified for clarity. In this sentence “presume” refers to the historical concept that direct target inhibition was solely responsible for antibiotic lethality. As the reviewer notes, there is now significant literature that ROS (and perhaps other secondary effects) also contribute to bacterial killing.

Line 73 However and the following might also: this phrasing, plus the presumed, misleads the reader from your intent. I suggest rephrasing.

See above re: line 72

Line 75 citations: these are inappropriate and should be changed to fit the statement. I suggest the initial paper by Collins (Kohanski 2007 Cell) a recent paper by Zhao (Zeng PNAS 2022), and a review Drlica Expert Rev Anti-infect Therapy 2021). The present

citations are fine if you want to narrow the statement to mycobacteria, but the history is that the E. coli work came first and was then generalized to mycobacteria. A mycobacterial paper for ROS is Shee 2022 AAC.

We thank the reviewer for noticing that we inadvertently omitted several important E. coli-related references. These have been added.

Line 75 and 76: Conversely ... unresolved. Compelling arguments have been made that show major flaws in the two papers cited, and a large body of evidence has now accumulated showing the validity of the idea promoted by the Collins lab, beginning with Kohanski 2007. In addition to many papers by Collins, see Hong 2019 PNAS and Zeng 2022 PNAS). It is fine if you want to counter the arguments against the Lewis and Imlay papers (summarized in Drlica & Zhao 2021 Expert Rev Anti-infect Therapy), but making a blanket statement suggests that the authors are unfamiliar with the literature.

We agree with the reviewer that the weight of the evidence supports a role for antibiotic-induced ROS as an important mechanism for antibiotic lethality under many (though not all) conditions. We have revised the text to better reflect this nuance.

Line 78. Advantages over what?

Text modified

Line 80 exposure: to finish the logic you need to show that E. coli and S. aureus persists fail to do this.

We thank the reviewer for their suggestion but studying these other organisms is outside the scope of this study.

Line 82 whereas: this misdirects the reader. It would seem that a simple "and" is better

Text modified

Line 89 I think this paragraph is about the need to study Mabs, the subject of the present report. This paragraph could use a more appropriate topic sentence to guide the reader so that no guessing is involved. I suggest rephrasing this paragraph to make the case for studying more compelling.

Text modified

Line 96. I suggest citing several references after subinhibitory concentration of antibiotic.

The references are in the following sentence alongside the key observations.

Line 99. Genetic analysis: how does this phrase fit with the idea of persister cells arising stochastically?

There are two issues: 1) We would argue that persister formation is not completely stochastic, but rather a probability that can be modified both genetically and by environment (for example hipA PMID: 6348026). 2) Even if persister formation were totally stochastic, the survival of these cells may depend on specific genes – as we indeed find in our Tn-Seq analysis of Mabs.

Line 106. In this paragraph you need to define persister. The consensus definition (Balaban 2019 Nat Micro Rev) is a subpopulation of tolerant cells. Tolerance is defined as the slowing or absence of killing while an antibiotic retains its ability to block growth. See Zeng 2022 PNAS for example with rapidly growing cells. Phenotypic tolerance is the absence of killing due to environmental perturbations, most notably nutrient starvation, dormancy, and growth to stationary phase. By extension, phenotypic persistence would be subpopulation status of a phenotypically tolerant cells. If you have a different definition, it is important to state it and emphasize that you disagree with the consensus statement.

Text modified

Line 109 unexpectedly. I would delete this word, because the literature leads the reader to expect this result unless you make a clear case for Mabs being fundamentally different from other bacteria with respect to how antibiotics kill bacteria (this is unlikely, see Shee 2022 AAC). Indeed, lines 111-113 state extensions of E. coli work, although suppression of ROS in phenotypic tolerance and genetic persistence have not been demonstrated.

Text modified

Line 124 you might add, in parentheses and with references, that a property of persisters is crosspersistence to multiple antibiotic classes. This is also true for tolerance, both genetic and phenotypic. An addition will support your approach.

Text modified

Line 128 minimal

Text not modified. We appreciate the reviewer's preference but both "minimal" and "minimum" are both widely accepted terms. Indeed, the Balaban et al 2019 consensus statement on definitions cited by the author above also uses "minimum" (PMID: 30980069), as do IDSA clinical guidelines (PMID: 39108079).

Line 130 is MIC somehow connected to killing or did you also measure killing? Note that blocking growth and killing cells are mechanistically distinct phenomena, although they are related. By being upstream from killing, blockage of growth will also interfere with killing.

Text modified

Line 133 PBS is undefined

Text modified

Line 134 increase in persisters ... you need to establish that these are not phenotypically tolerant cells. Do they constitute the entire population (tolerance)? Your data would be more indicative of persisters if you saw a distinct plateau with the PBS samples, as such data are often used to document persistence (retardation of killing is a property of tolerance, Balaban 2019). Fig. 1B is clearly phenotypic tolerance, as the entire population grows. Your data suggest that you are not measuring persistence as defined in the literature (Balaban 2019). Line 139 persister should be tolerance •

Text modified

Lines 142, 143, 144. 159, 163, 171, 181, 211, 226, 238, 246, 277, 279, 289 persistent should be tolerant

Text modified

Line 146 fig 1E Mtb does not show the adaptation phenomenon and it is clearly tolerant, not persistent. This should be pointed out. As stated, you may be misleading the reader.

Text modified

**Line 169. Please make it clear whether these genes are affecting antibiotic susceptibility (MIC will affect killing because blocking growth is upstream) or if you are dealing with tolerance (no change in MIC). These measurements are essential and should included as a table. By antibiotic response, do you mean that antibiotics change expression levels?*

Regarding MICs, the data for MICs in control and katG mutant are presented in Figure 4C and 4F. Regarding ‘response’ we have clarified the text of this sentence.

Line 174 Interestingly should be as expected

Text not modified; tetracyclines do not induce ROS in E. coli and oxazolidinones have not been studied in this regard.

Line 183 you need to include citations. You can cite the ability of chloramphenicol to block ROS-mediated killing of E. coli. That allows you to use the word unexpected

Text modified

Line 199. All of the data in Fig. 3 shows tolerance, not persistence, requiring word changes in this paragraph.

Text modified

Line 226. The MIC experiment is important. You can add that this result solidifies the idea that blocking growth and killing cells are distinct phenomena. You can cite Shee 2022 AAC for a mycobacterial paper

Text modified

Line 241. The result with levofloxacin is unexpected, because the fluoroquinolones are widely reported to induce ROS, even with mycobacteria (see Shee 2022 AAC). You need to point this out and perhaps redo the experiment to make sure it is correct.

We appreciate the reviewer’s interest in this question. All experiments in this paper were repeated multiple times. This particular experiment was repeated 3 times and in all replicates the katG mutant was sensitized to translation inhibitors but not levofloxacin. Shee et al examined Mtb treated with moxifloxacin and found ROS generation, but did not assess whether a Mtb katG mutant had impaired survival. Thus, in addition to differences in: i) the species studied and ii) the particular fluoroquinolone used, the two sets of experiments were designed to address different questions (ROS accumulation vs protection by katG) . A cell might accumulate ROS without a katG mutant having impaired survival if genetic redundancy exists – a result we indeed see in our clinical Mabs strains under some conditions (new data included in revised Figure 6A).

Line 269 Additional controls would bolster the conclusion: use of an antioxidant such as thiourea and an iron chelator (dipyridyl) both should reduce ROS effects.

New experiments performed, revised Figure 5.

Line 276 the word no is singular

Text modified

Line 284 this suggested ... in fact previous work suggested. This summary paragraph might go better as the first paragraph of the Discussion

Text modified to specify that this is in reference to the work in this manuscript

Lines 294-299 Most of this is redundant and should be deleted.

Text modified

Line 299 this species is vague

Text modified

Line 310 Do you want to discuss spoT?

Text not modified

Line 313 paragraph is largely redundant

Text modified

Line 314 controversial. As above, I would delete this, especially since it is not referenced and is unlikely to be true. If you believe it, you have the obligation to show why the ROS-lethality idea is untrue. If you are referring to Lewis and Imlay, there were almost a dozen supporting papers before 2013 and many after. This statement does not make the present work more important, so deletion costs you nothing.

Text modified

Line 314 direct disruption of targets. This is clearly not a general principle, because the quinolones rapidly kill while inhibition of gyrase by temperature-sensitive mutations does not (Kreuzer 1979 J.Bact; Steck 1985). Indeed, formation of drug-gyrase-DNA complexes is reversible: death is not.

Text modified

Line 318 as pointed out above, you have not brought this story up to date. The two papers mainly focused on Kohanski 2007, ignoring other available evidence."

Text modified

Line 326 you need to cite Shee 2022 AAC

Text modified

Line 342 the idea of mutants being protective is not novel, as several have been reported with E. coli studies. Thus, there is a general principle involved.

We agree that this suggests a potential general principle

Line 344. It depends on the inhibitor. For example, aminoglycosides are translation inhibitors and they also cause the accumulation of ROS.

We agree that ROS generation depends on the inhibitor, and indeed upon other variables including drug concentration, growth conditions, and bacterial species as well.

Line 347. You need to point out the considerable data showing that the absence of catalase increases killing

Text modified

Line 363 look at Shee 2022 AAC and Jacobs 2021 AAC

Text modified, reference added.

Line 585 I suggest having a colleague provide critical comments on the manuscript and acknowledge that person.

Text not modified

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